UNIT 4 – PROBABILITY

Probability theory provides a mathematical foundation to conceptssuch as “probability”, “information”, “belief”, “uncertainty”, “confidence”, “randomness”, “variability”, “chance” and “risk”. Probability theory is important to empirical scientists because it gives them a rational framework to make inferences and test hypotheses based on uncertain empirical data. Probability theory is also useful to engineers building systems that have to operate intelligently in an uncertain world. For example, some of the most successful approaches in machine perception (e.g., automatic speech recognition, computer vision) and artificial intelligence are based on probabilistic models. Moreover probability theory is also proving very valuable as a theoretical framework for scientists trying to understand how the brain works. Many computational neuroscientists think of the brain as a probabilistic computer built with unreliable components, i.e., neurons, and use probability theory as a guiding framework to understand the principles of computation used by the brain. Consider the following examples:

- WE need to decide whether a coin is loaded (i.e., whether it tends to favor one side over the other when tossed). WE toss the coin 6 times and in all cases WE get “Tails”. Would WE say that the coin is loaded?

- WE are trying to figure out whether newborn babies can distinguish green from red. To do so WE present two colored cards (one green, one red) to 6 newborn babies. WE make sure that the 2 cards have equal overall luminance so that they are indistinguishable if recorded by a black and white camera. The 6 babies are randomly divided into two groups. The first group gets the red card on the left visual field, and the second group on the right visual field. WE find that all 6 babies look longer to the red card than the green card. Would WE say that babies can distinguish red from green?

- A pregnancy test has a 99 % validity (i.e., 99 of of 100 pregnant women test positive) and 95 % specificity (i.e., 95 out of 100 non pregnant women test negative). A woman believes she has a 10 % chance of being pregnant. She takes the test and tests positive. How should she combine her prior beliefs with the results of the test?

- WE need to design a system that detects a sinusoidal tone of 1000Hz in the presence of white noise. How should design the system to solve this task optimally?

- How should the photo receptors in the human retina be interconnected to maximize information transmission to the brain?

Addition Rule:

Probability is defined as the total number of outcomes of an event that is associated with an experiment divided by the total number of outcomes that are possible as a result of that experiment. “n (S)” denotes the number of outcomes of the experiment where ‘S’ is the sample space that is a set of all possible outcomes and A an event whose number of outcomes is given by n (A).

Then, the probability of occurrence of the event A, $P(A) = \frac{n(A)}{n(S)}$

Addition rules are important in probability. These rules provide us with a way to calculate the probability of the event "*A* or *B*", provided that we know the probability of *A* and the probability of *B*. Sometimes the "or" is replaced by U, the symbol from set theory that denotes the union of two sets. The precise addition rule to use is dependent upon whether event *A* and event *B* are mutually exclusive or not.

Addition Rule for Mutually Exclusive Events

If events *A* and *B* are mutually exclusive, then the probability of *A* or *B* is the sum of the probability of *A* and the probability of *B*. We write this compactly as follows:

$P(A \text{ or } B) = P(A) + P(B)$

The above formula can be generalized for situations where events may not necessarily be mutually exclusive. For any two events *A* and *B*, the probability of *A* or *B* is the sum of the probability of *A* and the probability of *B* minus the shared probability of both *A* and *B*:

$P(A \text{ or } B)=P(A)+P(B)-P(A \text{ and } B)$

or

$P(A\cup B)=P(A)+P(B)-P(A\cap B)$

If A and B are two mutually exclusive events , $P(A\cap B)=0$. Then the probability that either one of the events will occur is:

$P(A\cup B)=P(A)+P(B)$

We will see examples of how to use these addition rules.

Suppose that we draw a card from a well shuffled standard deck of cards. We want to determine the probability that the card drawn is a two or a face card. The event "a face card is drawn" is mutually exclusive with the event "a two is drawn." So we will simply need to add the probabilities of these two events together.

There are a total of 12 face cards, and so the probability of drawing a face card is 12/52. There are four twos in the deck, and so the probability of drawing a two is 4/52. This means that the probability of drawing a two or a face card is 12/52 + 4/52 = 16/52

Now suppose that we draw a card from a well shuffled standard deck of cards. Now we want to determine the probability of drawing a red card or an ace. In this case the two events are not mutually exclusive. The ace of hearts and the ace of diamonds are elements of the set of red cards and the set of aces.

We consider three probabilities and then combine them using the generalized addition rule.

- The probability of drawing a red card is 26/52
- The probability of drawing an ace is 4/52
- The probability of drawing a red card and an ace is 2/52

This means that the probability of drawing a red card or an ace is:

26/52+4/52 - 2/52 = 28/52.

Multiplication Rule of Probability

Multiplication rule probability (General)

The **multiplication rule** is a way to find the probability of two events happening at the same time (this is also one of the AP Statistics formulas). There are two multiplication rules. The general multiplication rule formula is: $P(A \cap B) = P(A) P(B|A)$ and the specific multiplication rule is $P(A \text{ and } B) = P(A) * P(B)$. $P(B|A)$ means "the probability of A happening given that B has occurred".

Multiplication Rule Probability (Specific)

The **specific multiplication rule**, $P(A \text{ and } B) = P(A) * P(B)$, is *only* valid if the two events are independent. In other words, it only works if one event does not change the probability of the other event.
Examples of independent events :

- Owning a cat and getting a weekly paycheck.
- Finding a parking space and having a coin for the meter.
- Buying a book and then buying a coffee.

Multiplication Rule Probability: Using the Specific Rule Example:

Using the specific multiplication rule formula is very straightforward. Just multiply the probability of the first event by the second. For example, if the probability of event A is 2/9 and the probability of event B is 3/9 then the probability of both events happening at the same time is (2/9)*(3/9) = 6/81 = 2/27.

Multiplication Rule Probability: Using the General Rule

This rule can be used for any event (they can be independent or dependent events). You still have to multiply two numbers, but first you have to use a little logic to figure out the second probability before multiplying.

Sample problem: a bag contains 6 black marbles and 4 blue marbles. Two marbles are drawn from the bag, without replacement. What is the probability that both marbles are blue?

Step 1: Label your events A and B. Let A be the event that marble 1 is blue and let B be the event that marble 2 is blue.

Step 2: Figure out the probability of A. There are ten marbles in the bag, so the probability of drawing a blue marble is 4/10.

Step 3: Figure out the probability of B. There are nine marbles in the bag, so the probability of choosing a blue marble (P B|)A is 3/9.

Step 4: Multiply Step 2 and 3 together: (4/10)*(3/9) = 2/15.

Conditional Probability

A **conditional probability** is a probability that a certain event will occur given some knowledge about the outcome or some other event.

The formula for conditional probability is:

P(B|A) = P(A and B) / P(A)

which can also be rewritten as:

P(B|A) = P(A∩B) / P(A)

Example 1. In a group of 100 sports car buyers, 40 bought alarm systems, 30 purchased bucket seats, and 20 purchased an alarm system and bucket seats. If a car buyer chosen at random bought an alarm system, what is the probability they also bought bucket seats?

Step 1: Figure out P(A). It's given in the question as 40%, or 0.4.

Step 2: Figure out P(A∩B). This is the intersection of A and B: both happening together. It's given in the question 20 out of 100 buyers, or 0.2.

Step 3: Insert your answers into the formula:
P(B|A) = P(A∩B) / P(A) = 0.2 / 0.4 = 0.5.
The probability that a buyer bought bucket seats, given that they purchased an alarm system, is 50%.

Conditional Probability in Real Life

Conditional probability is used in many areas, including finance, insurance and politics. For example, the re-election of a president depends upon the voting preference of voters and perhaps the success of television advertising — even the probability of the opponent making gaffes during debates!

The weatherman might state that your area has a probability of rain of 40 percent. However, this fact is *conditional* on many things:

- The probability of a cold front coming to your area.
- The probability of rain clouds forming.
- The probability of another front pushing the rain clouds away.

We say that the **conditional probability** of rain occurring depends on all the above events.

Where does the Conditional Probability Formula Come From?

The formula for conditional probability is derived from the probability multiplication rule, P(A and B) = P(A)*P(B|A). You may also see this rule as P(A∪B). The Union symbol (∪) means "and", as in event A happening and event B happening. Step by step, here's how to derive the conditional probability equation from the multiplication rule:

Step 1: Write out the multiplication rule:
P(A and B) = P(A)*P(B|A)

Step 2: Divide both sides of the equation by P(A):
P(A and B) / P(A) = P(A)*P(B|A) / / P(A)

Step 3: Cancel P(A) on the right side of the equation:
P(A and B) / P(A) = P(B|A)

Step 4: Rewrite the equation:
P(B|A) = P(A and B) / P(A)

Bayes' Theorem:

Bayes' theorem is a way to figure out conditional probability. Conditional probability is the probability of an event happening, given that it has some relationship to one or more other events. For example, your probability of getting a parking space is connected to the time of day you park, where you park, and what conventions are going on at any time. Bayes' theorem is slightly more nuanced. In a nutshell, it gives you the actual probability of an **event** given information about **tests**.

- "Events" Are different from "tests." For example, there is a **test** for liver disease, but that's separate from the **event** of actually having liver disease.

- **Tests are flawed**: just because you have a positive test does not mean you actually have the disease. Many tests have a high false positive rate. **Rare events tend to have higher false positive rates** than more common events. We're not just talking about medical tests here. For example, spam filtering can have high false positive rates. Bayes' theorem takes the test results and calculates your *real probability* that the test has identified the event.

Bayes' Theorem (also known as Bayes' rule) is a deceptively simple formula used to calculate conditional probability. The Theorem was named after English mathematician Thomas Bayes (1701-1761). The formal definition for the rule is:

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

In most cases, you can't just plug numbers into an equation; You have to figure out what your "tests" and "events" are first. For two events, A and B, Bayes' theorem allows you to figure out p(A|B) (the probability that event A happened, given that test B was positive) from p(B|A) (the probability that test B happened, given that event A happened). It can be a little tricky to wrap your head around as technically you're working backwards; you may have to switch your tests and events around, which can get confusing. An example should clarify what I mean by "switch the tests and events around."

Bayes' Theorem Example

One might be interested in finding out a patient's probability of having liver disease if they are an alcoholic. "Being an alcoholic" is the **test** (kind of like a litmus test) for liver disease.

- **A** could mean the event "Patient has liver disease." Past data tells you that 10% of patients entering your clinic have liver disease. P(A) = 0.10.
- **B** could mean the litmus test that "Patient is an alcoholic." Five percent of the clinic's patients are alcoholics. P(B) = 0.05.
- You might also know that among those patients diagnosed with liver disease, 7% are alcoholics. This is your **B|A:** the probability that a patient is alcoholic, given that they have liver disease, is 7%.

Bayes' theorem tells you:

P(A|B) = (0.07 * 0.1)/0.05 = 0.14

In other words, if the patient is an alcoholic, their chances of having liver disease is 0.14 (14%). This is a large increase from the 10% suggested by past data. But it's still unlikely that any particular patient has liver disease.